



The Complete Platform Guide: What CyberTrace Is, How It Works, and Every Feature Explained With Real-World Examples

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Indian Cyber Crime Coordination Centre (I4C) • Ministry of Home Affairs, Government of India

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Global Benchmarks: Chainalysis, TRM, Arkham, Elliptic

Status: Production Ready v2.0

Target Focus: 1930 Helpline & I4C Police Officers

GitHub Repo: github.com/Hidayatulla268/CyberTrace

Live App: hidayatulla268.github.io/CyberTrace/

What is CyberTrace in Simple Words?

Think of CyberTrace as the **'Google Maps + Digital Sherlock Holmes'** for **Cryptocurrency Crime**. When an ordinary person is scammed online (e.g. a fake Telegram part-time job, investment fraud, or ransomware), the victims transfer money into a crypto wallet address given by the scammer. Because cryptocurrency transactions are public but anonymous, police officers previously had to spend weeks manually looking at blockchain explorers. **CyberTrace automates this entire investigation in 1 second:** it follows where the money went, identifies the scam syndicate, locates which exchange the money was deposited into, and writes the police legal notice to freeze the scammer's bank account.

THE 6-STEP MASTER FORENSIC WORKFLOW (USP)

1. REPORT
Victim Intake



2. TRACE
Stolen Funds



3. CONNECT
Hidden Wallets



4. IDENTIFY
Exchange CEX



5. MONITOR
Live Mempool



6. PROVE
I4C Dossier

1. Detailed Feature Guide with Practical Examples (Part 1)

■ 1. Smart Multi-Chain Wallet Scanner

What it does: Scans any wallet address (EVM, Tron, Bitcoin, Solana) and instantly displays total received funds, sent funds, transaction count, active dates, and live on-chain balances.

Real-World Example: An investigator enters suspect address 0xA1b2...9T0. The system displays ■8,42,000 received across 128 transactions, showing activity started on 12 Aug 2026 and last moved today at 05:42 PM.

■ 2. Stolen Money Tracker (Tranche Following)

What it does: Instead of just looking at total volume, it locks onto a specific victim's deposit (e.g. ■50,000) and follows that exact tranche hop-by-hop as the scammers split and launder it.

Real-World Example: Victim deposits ■50,000. CyberTrace calculates: Hop 1 split ■30k (60%) to Mule Wallet A and ■20k (40%) to Binance. Hop 2 split ■18k (36%) to WazirX and ■12k (24%) to an intermediary. Offers 1-click CSV export of the full money trail.

■■ 3. Animated Fund-Flow Graph

What it does: Visually diagrams the flow of money as an interactive flowchart: Victim → Scammer Hub → Splitter Mules → Centralized Exchange with animated particles showing the direction of speed.

Real-World Example: Instead of reading a confusing list of 100 transaction hashes, the police officer sees clear glowing boxes connected by arrows, showing exactly how funds moved from the victim's wallet to Binance Hot Cluster 14.

■ 4. Time-Travel Transaction Scrubber

What it does: An interactive playback bar on the Fund-Flow graph with Play, Pause, Step Back, and Speed (1x to 4x) controls that replays the crime chronologically from Hour 0 to Hour 24.

Real-World Example: Dragging the scrubber slider from 0% to 100% shows: at T+00:00 victim funds arrive; at T+00:25 the first split occurs; at T+00:27 funds hit the exchange deposit gateway, proving automated laundering in under 30 seconds.

■ 5. Exchange Finder (CEX Attribution)

What it does: Automatically detects whether stolen funds landed inside a centralized crypto exchange (such as Binance, WazirX, CoinDCX, KuCoin, or OKX) where KYC identity is stored.

Real-World Example: Analyzes the final recipient address and confirms with **91% confidence** that the address belongs to Binance Hot Cluster 14, enabling police to know exactly which company to subpoena.

■ 6. Hidden Wallet & Accomplice Detector

What it does: Uncovers accomplice wallets that never directly touched the victim, including Gas Sponsors (who paid transaction fees), Co-Spenders, and Peeling Splitters.

Real-World Example: Discovers that wallet 0xFeeF...119A funded gas fees for 12 different scammer hubs via FixedFloat, exposing the underlying bot infrastructure.

2. Detailed Feature Guide with Practical Examples (Part 2)

■ 7. Heuristic AI Risk Score Gauge (0–100)

What it does: Calculates a dynamic risk score from 0 (Safe) to 100 (Critical Risk) with clear bullet-pointed reasons explaining why the wallet is dangerous.

Real-World Example: *Suspect wallet gets an 87/100 HIGH RISK score because: (1) Rapid peeling chain detected, (2) Funds split across 3 wallets in 12 mins, (3) Consolidated into Binance sweep gateway, (4) Reported by multiple victims on 1930 portal.*

■ 8. Fraud Pattern Detection Engine

What it does: Detects automated laundering signatures including: Peeling Chains (shaving off small tranches), Rapid Transfers (<45s cadence), Fan-Out (splitting to multiple mules), and Fan-In (re-gathering funds).

Real-World Example: *Detects that funds were transferred across 3 hops in 42 seconds, proving the scam is operated by an automated bot network rather than a human user.*

■ 9. Fraud DNA™ Syndicate Attribution (Zero-Day Detection)

What it does: Detects new, unknown scam wallets even if nobody has ever reported them before, by matching their behavioral sequence fingerprint against known crime syndicate playbooks.

Real-World Example: *Scammers deploy brand new Wallet Z. CyberTrace calculates an 8-dimensional vector (timing, peeling ratio, exchange gateway) and alerts: '■ 91% Match to Campaign #CYB-2048 (Hydra-Peel Telegram Scam Syndicate)'.*

■ 10. Cross-Chain Bridge & Hop Tracker (TRM Labs Style)

What it does: Follows funds when scammers try to evade police by hopping across blockchains (e.g. from Ethereum to Tron, BSC, Bitcoin, or Solana) using bridge protocols like Across and FixedFloat.

Real-World Example: *Scammer deposits ERC-20 USDT on Ethereum, bridges via Across Protocol to TRC-20 USDT on Tron, and off-ramps to Binance BSC. CyberTrace unifies all chains into one continuous forensic flow.*

■ 11. 100k+ Entity & Deanonymization Directory (Arkham Style)

What it does: Searchable database of 100,000+ indexed real-world labels: Centralized Exchanges, Lazarus Group APT addresses, LockBit Ransomware vaults, and Darknet marketplaces.

Real-World Example: *Search for 'Lazarus' or 'Binance' to immediately see associated hotwallet clusters, historical stolen volume (■4,800 Cr), and click 'Trace' to load full transaction ledgers.*

■ 12. Mixer Demasking Engine (Elliptic Style)

What it does: De-anonymizes privacy pools like Tornado.Cash by identifying common relay gas dispatchers (e.g. 0xRelay99B) and matching deposit-withdrawal timing brackets.

Real-World Example: *Suspect deposits 100 ETH into Tornado Cash. CyberTrace links the gas relayer to the withdrawal wallet 32 minutes later, shrinking the anonymity set from 10,000 to a single target with 94.2% confidence.*

3. Detailed Feature Guide with Practical Examples (Part 3)

■ ■ 13. Global Sanctions & OFAC Screener

What it does: Real-time anti-money laundering (AML) screening checking whether any suspect wallet is listed on US OFAC SDN, UN Sanctions, EU Blacklists, or FIU-IND registries.

Real-World Example: Scanning a ransomware address instantly displays a red badge: '**OFAC SDN Sanctioned — DPRK State Sponsored Threat**', alerting police to involve international agencies.

■ ■ 14. Live Mempool Monitoring Stream

What it does: Live blockchain feed that watches flagged wallets in real-time and alerts investigators with visual pulse beacons the exact second new transactions are broadcast.

Real-World Example: While an officer is investigating Case #1245, a new ■80,000 transaction is broadcast to the mempool. The live feed lights up green: 'Live Event: ■80,000 moved to Binance Hot 14', enabling instant action.

■ 15. Investigator Case Workspace & Evidence Vault

What it does: Multi-case organizer where police officers can manage multiple FIRs, assign junior officers, write encrypted case notes, and maintain an audit log of evidence.

Real-World Example: Officer records notes: 'Requested Binance KYC for UID #8912401 on 24 Aug. Bank account frozen via 1930 portal.' Notes are encrypted and stored in the local vault.

■ 16. Multi-Jurisdictional Legal Subpoena Dispatcher

What it does: 1-click generator and emergency API dispatcher that formats complete legal notices under Section 91 CrPC (India), US CLOUD Act § 2703(d), and MLAT freeze orders.

Real-World Example: Automatically generates a formal Section 91 CrPC notice pre-filled with the transaction hash, deposit amount (■20,000), timestamp, and sends it directly to Binance's Nodal Officer.

■ 17. Public Wallet Safety Check ('Check Before You Send')

What it does: A public consumer protection tool where citizens can paste any wallet address before paying someone online to verify if it is safe or a reported scam.

Real-World Example: A student is asked to send ■10,000 to a crypto wallet for a 'Telegram job'. They check the address on CyberTrace, which displays: '■ DO NOT SEND — 91% Match with Known Fraud Syndicate', preventing the loss before it happens.

■ 18. Real-Time On-Chain RPC Querying (Live Web3 Mode)

What it does: Directly queries live Ethereum and BSC mainnet RPCs in real-time, fetching real on-chain ETH balances, nonces, and transaction counts for any live wallet.

Real-World Example: Entering Vitalik Buterin's address (0xd8dA...6045) queries Ethereum mainnet and displays real live balance: **6.6410 ETH (■18,26,275)** across 5,956 live transactions.

4. Global Benchmark Comparison (Chainalysis vs. CyberTrace)

Capability / Feature	Chainalysis	TRM Labs	Arkham	CyberTrace (Ours)
Fund-Flow Graph	Yes	Yes	Yes	Yes (Interactive SVG)
Time-Travel Scrubber	No	No	No	Yes (Play/Pause/Speed)
Cross-Chain Bridges	Yes	Yes	No	Yes (Across/FixedFloat)
Entity Directory	Yes	Yes	Yes	Yes (100k+ Indexed)
Mixer Demasking	Partial	Partial	No	Yes (Relayer Analysis)
Fraud DNA™ (Zero-Day)	No	No	No	Yes (8-D Vector Match)
Section 91 CrPC Notice	No	No	No	Yes (Auto-Drafted)
Citizen Safety Screener	No	No	No	Yes (Free Consumer Tool)
Price / License	\$50k+/year	\$60k+/year	Freemium	Free for Police/LEAs

5. Step-by-Step 7-Stage Guided Live Demo Tour

Step 1: Enter Address	Click the preset pill Task-Based Telegram Scam (■50k) to load suspect address 0xA1b2...9T0.
Step 2: Trigger Analysis	Click Analyze . The automated engine scans transactions, queries balances, and identifies risk patterns.
Step 3: Risk Score	Observe the 87/100 High Risk AI gauge and the 5-point evidence checklist.
Step 4: Flow Graph	Play the Time-Travel Scrubber to see money move chronologically from Victim to Binance.
Step 5: Exchange Detection	See Binance Hot Cluster 14 detected with 91% confidence.
Step 6: Fraud DNA™ Zero-Day	Click ■ Unreported Wallet Z to see zero-day detection matching Campaign #CYB-2048 at 91%.
Step 7: Generate Dossier	Click Download Dossier to export the official Section 91 CrPC court evidence PDF.

6. Technology Architecture & System Design

Zero-Dependency Client-Side Engine:

Built using pure semantic HTML5, Vanilla JavaScript (ES6+), and responsive CSS3. Requires zero npm builds or server installs, allowing it to run instantly on any police department workstation, tablet, or forensic laptop even in air-gapped or restricted intranet environments.

Live Web3 RPC Layer:

Connects to decentralized public JSON-RPC nodes on Ethereum, BSC, and Polygon to query real balances and nonces dynamically.

7. Quick Reference Links & Contacts

Live Platform (GitHub Pages):	https://hidayatulla268.github.io/CyberTrace/
GitHub Repository:	https://github.com/Hidayatulla268/CyberTrace
Documentation PDF File:	CyberTrace_Platform_Guide.pdf
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